

MATH3303 – Lecture 25

Prime Ideals; Introduction to Maximal Ideals

Review Notes

Reference: Lee, Abstract Algebra, Section 9.6

Overview

This lecture introduces *prime ideals* – the first of two distinguished classes of ideals – and begins the discussion of *maximal ideals*.

- **Definition of a prime ideal** – generalising the divisibility property of prime integers
- $\{0\}$ is prime $\iff R$ is an integral domain
- **Prime ideals in $\mathbb{Z}$** are precisely (0) and (p) for p prime
- **Quotient characterisation:** I prime $\iff R/I$ is an integral domain
- **Examples in $\mathbb{Z}[x]$:** both (x) and (x, p) are prime, via the First Isomorphism Theorem
- **Maximal ideals:** definition and the example of $(p) \trianglelefteq \mathbb{Z}$

Connection to Lecture 24. The First Isomorphism Theorem from Lecture 24 lets us identify quotient rings R/I with images of ring homomorphisms. The quotient characterisation in this lecture (Theorem 2.1) builds directly on that: to show I is prime, we exhibit R/I as an integral domain, often by realising it as the image of an evaluation map.

1 Prime Ideals: Definition and First Examples

Definition 1.1: Prime ideal (from lecture; cf. Lee, Def. 9.10)

Let R be a commutative ring and I an ideal of R . Then I is said to be *prime* if

- (1) $I \neq R$; and
- (2) for all $x, y \in R$, $xy \in I \implies x \in I$ or $y \in I$.

Remark 1.1: Note on noncommutative rings (from lecture)

For noncommutative rings the definition of a prime ideal is different. In this course we only consider prime ideals in commutative rings.

Example 1.1: Zero ideal and integral domains (from lecture; cf. Lee, Example 9.32)

Let R be a commutative ring. Then

$$R \text{ is an integral domain} \iff \{0\} \text{ is a prime ideal of } R.$$

Proof.

$$\begin{aligned} R \text{ is an integral domain} &\iff R \text{ has no zero divisors} \\ &\iff ab = 0 \implies a = 0 \text{ or } b = 0 \\ &\iff \{0\} \text{ is a prime ideal.} \quad \square \end{aligned}$$

In particular, $\{0\}$ is a prime ideal in $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{Z}[x]$, and $\mathbb{Q}[x]$, since each is an integral domain.

Example 1.2: Prime ideals in $\mathbb{Z}$ (from lecture; cf. Lee, Example 9.33)

In $\mathbb{Z}$, the ideal $(n) = \{na : a \in \mathbb{Z}\}$ (where $n \geq 0$) is prime if and only if $n = 0$ or n is a prime number.

($\Rightarrow$) Suppose (n) is prime, $n \geq 0$.

- Note $n \neq 1$, otherwise $(n) = \mathbb{Z}$, contradicting the definition of a prime ideal.
- Suppose for contradiction $n \neq 0$ and n is not prime. Then $n = xy$ with $x, y \neq 1$ and $x, y \neq 0$. So $xy \in (n)$ but neither x nor y lies in (n) , a contradiction.

Hence $n = 0$ or n is prime.

($\Leftarrow$) If $n = 0$: since $\mathbb{Z}$ is an integral domain, (0) is prime by Example 1.2.

If n is prime, then for any $xy \in (n)$, by definition $n \mid xy$, so $n \mid x$ or $n \mid y$, i.e. $x \in (n)$ or $y \in (n)$. Hence (n) is prime.

Exercise 1.1: Prime ideals in $\mathbb{C}[x]$ (from lecture)

Determine all prime ideals in $\mathbb{C}[x]$.

2 The Quotient Characterisation

Theorem 2.1: Quotient test for prime ideals (from lecture; cf. Lee, Thm. 9.21)

Let R be a commutative ring and I an ideal of R . Then

$$I \text{ is a prime ideal} \iff R/I \text{ is an integral domain.}$$

Proof. Recall that the zero element of R/I is $I = 0 + I$.

($\Rightarrow$) Let I be prime and suppose $(r + I)(s + I) = I$ in R/I , where $r, s \in R$. Then $rs + I = I$, so $rs \in I$. Since I is prime, $r \in I$ or $s \in I$, i.e. $r + I = I$ or $s + I = I$. So R/I has no zero divisors. Hence R/I is an integral domain.

($\Leftarrow$) Let R/I be an integral domain and suppose $rs \in I$. Then $I = rs + I = (r + I)(s + I)$. Since R/I has no zero divisors, $r + I = I$ or $s + I = I$, hence $r \in I$ or $s \in I$. So I is prime. $\square$

3 Two Key Examples in $\mathbb{Z}[x]$

Example 3.1: (x) is prime in $\mathbb{Z}[x]$ (from lecture; cf. Lee, Example 9.35)

Consider the evaluation-at-zero map

$$\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}, \quad f(x) \mapsto f(0).$$

Then φ is a surjective ring homomorphism with $\ker \varphi = (x)$. By the First Isomorphism Theorem,

$$\mathbb{Z}[x]/(x) \cong \mathbb{Z}.$$

Since $\mathbb{Z}$ is an integral domain, (x) is a prime ideal of $\mathbb{Z}[x]$ by Theorem 2.1.

Example 3.2: (x, p) is prime in $\mathbb{Z}[x]$ for p prime (from lecture)

Let p be a prime number. Define

$$\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}_p, \quad f(x) \mapsto f(0) \bmod p.$$

This is the composition $\mathbb{Z}[x] \xrightarrow{\text{eval at } 0} \mathbb{Z} \xrightarrow{\text{mod } p} \mathbb{Z}_p$, hence a ring homomorphism. It is clearly surjective.

Exercise. Check $\ker \varphi = (x, p)$.

($\supseteq$) Both x and p map to 0 under φ , so $(x, p) \subseteq \ker \varphi$.

($\subseteq$) Let $f(x) \in \ker \varphi$, so $f(0) \equiv 0 \pmod{p}$. Write $f(x) = f(0) + xg(x)$ for some $g(x) \in \mathbb{Z}[x]$. Since $p \mid f(0)$, we have $f(0) \in (p) \subseteq (x, p)$, and $xg(x) \in (x) \subseteq (x, p)$. Hence $f(x) \in (x, p)$.

Apply FIT. By the First Isomorphism Theorem,

$$\mathbb{Z}[x]/(x, p) \cong \mathbb{Z}_p.$$

Since $\mathbb{Z}_p$ is a field, hence an integral domain, (x, p) is a prime ideal of $\mathbb{Z}[x]$ by Theorem 2.1.

4 Maximal Ideals: Definition and the Example of $\mathbb{Z}$

Definition 4.1: Maximal ideal (from lecture; cf. Lee, Def. 9.9)

Let R be a commutative ring and $I \leq R$. Then I is said to be *maximal* if

- (1) $I \neq R$ (i.e. I is a *proper* ideal); and
- (2) there is no ideal J with $I \subsetneq J \subsetneq R$.

Equivalently: whenever $I \subseteq J \subseteq R$ with J an ideal, either $J = I$ or $J = R$.

Example 4.1: All maximal ideals in $\mathbb{Z}$ (from lecture; cf. Lee, Example 9.27)

The maximal ideals in $\mathbb{Z}$ are exactly (p) for p a prime.

Every ideal of $\mathbb{Z}$ is principal, i.e. of the form (n) for some $n \in \mathbb{Z}_{\geq 0}$.

Claim. (n) is maximal $\iff n$ is prime.

($\Rightarrow$) Suppose for contradiction n is composite, say $n = kl$ with $1 < k, l < n$. Then $n \in (k)$, so $(n) \subseteq (k) \subsetneq \mathbb{Z}$. In fact $(n) \subsetneq (k)$, contradicting maximality.

($\Leftarrow$) Suppose p is prime. We show (p) is maximal.

- $(p) \neq \mathbb{Z}$, since $p \geq 2$.
- Let $(p) \subseteq (n) \subseteq \mathbb{Z}$. Then $p \in (n)$, so $n \mid p$. Hence $n = 1$ or $n = p$ (up to sign), giving $(n) = \mathbb{Z}$ or $(n) = (p)$.

So (p) is maximal. □

Remark 4.1: Looking ahead (from lecture)

The next lecture develops a general test for maximality (the $(x, I) = R$ criterion) and the parallel quotient characterisation: I is maximal $\iff R/I$ is a field. This will let us recognise maximal ideals in $\mathbb{C}[x]$, $\mathbb{C}[x, y]$, and other rings.

Checklist of Skills

- ☐ State the definition of a prime ideal in a commutative ring.
- ☐ Prove that $\{0\}$ is prime iff R is an integral domain.
- ☐ Classify all prime ideals of $\mathbb{Z}$: namely (0) and (p) for p prime.
- ☐ State and prove the quotient test: I prime $\iff R/I$ is an integral domain.
- ☐ Use the First Isomorphism Theorem to verify (x) is prime in $\mathbb{Z}[x]$ via $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$.
- ☐ Compute $\ker(\mathbb{Z}[x] \rightarrow \mathbb{Z}_p, f \mapsto f(0) \bmod p)$ and conclude (x, p) is prime.
- ☐ State the definition of a maximal ideal.
- ☐ Prove that the maximal ideals of $\mathbb{Z}$ are exactly (p) for p prime.